

Department of Statistics and Computer Science
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CSC2021 – Programming using Data Structures
Labsheet 06

Binary Search Trees (BST)

Purpose

- Understanding the implementation of Binary Search Tree (BST) data structure and its components.
- Understand the tree traversing algorithms.

Theory Recursion (A brief reminder)

- A recursive program is a one that calls itself.
- It should,
 - Have a base case or a *termination condition*.
 - Change its state and move towards the termination condition.
 - Call itself, recursively

Binary Search Trees (BST) - An Overview

- A non-Linear Data Structure.
- A tree (upside-down) is an abstract model of a hierarchical structure
- Each node has at most two descendants.
- Each element except the topmost element has a ***parent*** and zero or more elements which are called ***children***.
- In Computer Science, the word tree is mentioned to describe rooted trees.
- A ***rooted tree*** is a tree with one of its nodes (top most node) declared as the root of the tree (using a pointer).
- The two children nodes are called **left** child node and **right** child node.
- Nodes with no children are called ***terminal nodes or leaf nodes***.
- In BST the **left child** < **parent** and **right child** > **parent**.
- Equal elements can be placed as left child or right child. However it should be consistent throughout the tree. (Should be placed in the same side throughout)
- Some common terminology,
 - Levels of a tree - The horizontal levels of which the nodes of multiple branches are aligned.
 - Level zero - root node (level)
 - Height of a node - Maximum number of levels/ edges from that node to its deepest leaf node
 - Height of a tree - Maximum number of levels/ edges from root to the deepest leaf node.
- Height of a tree
- A Basic BST
- Inserting into a binary search tree (Insertion of 19)
- Basic structure of a binary search tree has

- A **Node** which contains a data part as well as two references of **Node** type in order to handle the left (**left**) child and the right child (**right**)
- A BST class which contains a reference to the root node and several methods that can be used to manipulate the binary search tree.

- An method for inserting new nodes to a tree using existing nodes in the tree (Uses recursion to find the place that the node is to be inserted).
- A searchLevel method that is used to search a given tree for the level at which a given value is found first.

Question 01

1. Create a recursive method to find the factorial of any given number

Eg : $5! = 5*4*3*2*1 = 120$

2. Create a recursive method that can be used to print the fibonacci series of any given length

Eg : Fibonacci series of length 10 - **1 1 2 3 5 8 13 21 34 55**

3. A thermometer has recorded the following values of temperature. Create a Binary Search Tree data structure and respective methods mentioned above in order to store those values.

- a. 44, 27, 37, 60, 12, 28, 62, 45
- b. Search the tree for the level of the node containing value 12.